

An extension of Khovanov homology to singular links and tangles

Isaiah Martinez
Advisor: Carmen Caprau

California State University, Fresno

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Outline

- 1 Introduction
- 2 The polynomial invariant
- 3 Categorification

Introduction

Singular Crossings

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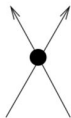


Figure: A singular crossing.

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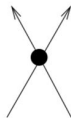


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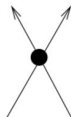
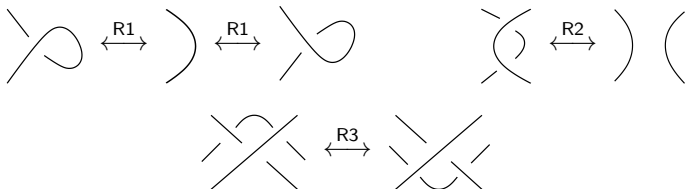


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- Recall that we typically don't allow our links to self-intersect (they are embeddings), but singular links are allowed to have finitely many such points of self-intersection.
- Singular link diagrams can have classical crossings just as regular links, and they can also have finitely many singular crossings.

The Reidemeister (type) moves on singular link diagrams



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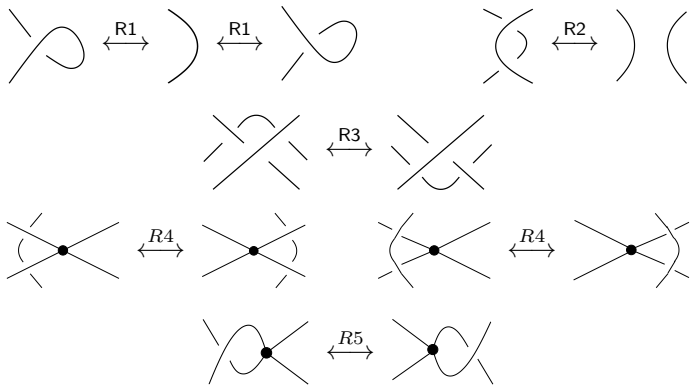


Figure: The Reidemeister (type) moves

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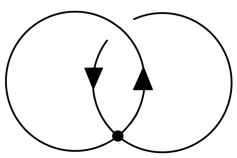


Figure: An oriented singular link diagram

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- That is, we will construct a chain complex associated to each singular link diagram so that
 - the chain complex is an oriented singular link invariant and
 - after applying a degree preserving functor, the graded Euler characteristic of the complex recovers the polynomial $\overline{P}_L(q)$. That is,

$$\overline{P}_L(q) = \sum_{i,j \in \mathbb{Z}} (-1)^i q^j \text{rank}(H^{i,j}(L))$$

The polynomial invariant

Resolving crossings in diagrams

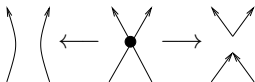
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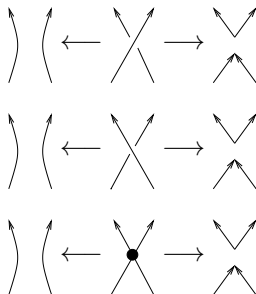
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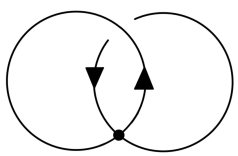


- After resolving all the crossings of a fixed diagram D , we call the resulting web Γ a *state* of the diagram D .

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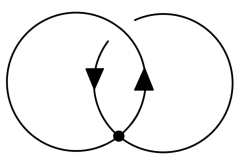
The four states of the singular link



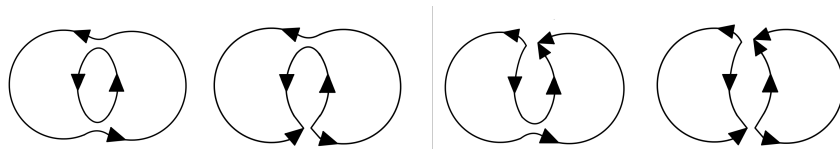
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- This sends an oriented link diagram to a $\mathbb{Z}[q, q^{-1}]$ -linear combination of oriented planar graphs with bivalent vertices, called *webs*.

Web skein relations and the $sl(2)$ polynomial

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- Webs are evaluated via the relations:

$$\langle \bigcirc \cup \Gamma \rangle = (q + q^{-1}) \langle \Gamma \rangle = \langle \bigcirc \cup \Gamma \rangle$$

$$\langle \text{down-left} \rightarrow \text{down-right} \rangle = \langle \text{down-left} \rightarrow \text{down-right} \rangle \quad \text{and} \quad \langle \text{down-left} \leftarrow \text{down-right} \rangle = \langle \text{down-left} \leftarrow \text{down-right} \rangle$$

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- The $sl(2)$ polynomial can then be defined as

$$P_D(q) := \sum_{\Gamma} \pm q^{\alpha(\Gamma)} \langle \Gamma \rangle \in \mathbb{Z}[q, q^{-1}],$$

where the sum is over all states/webs Γ associated with an oriented link diagram D , and the exponents $\alpha(\Gamma)$ and the sign $\pm$ are determined by the crossing decomposition rules.

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$$\langle \text{down-left to up-right} \rangle = \langle \text{curved arrow} \rangle \quad \text{and} \quad \langle \text{up-left to down-right} \rangle = \langle \text{curved arrow} \rangle$$

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- If $D_1 \stackrel{\text{Reid.}}{\sim} D_2$, then $P_{D_1}(q) = P_{D_2}(q)$, and $P_L(q) := P_D(q)$ is the $sl(2)$ polynomial, or equivalently, the *unnormalized Jones polynomial*.

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- With this and the same skein relations as before, we construct our polynomial

$$\overline{P}_L(q) = \sum_{\Gamma} \pm q^{\alpha(\Gamma)} \langle \Gamma \rangle$$

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Theorem

$\overline{P}_L(q)$ is an oriented singular link invariant. That is, if $D_1 \stackrel{Reid.}{\sim} D_2$, then $\overline{P}_{D_1}(q) = \overline{P}_{D_2}(q)$.

Categorification

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- The chain complex $[[T]]$ is constructed as a formal tensor product of these complexes over all states of the tangle T .

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- The differential D in the complex for the singular crossing on the last slide is the diagonal matrix of foams

$$D = \begin{bmatrix} \text{foam} & 0 \\ 0 & \text{foam} \end{bmatrix}.$$



The diagram shows a 2D projection of a 3D foam structure. It consists of several planar faces meeting at a central point. A red curve is drawn on one of the faces, and arrows indicate the orientation of the faces and the curve. The foam is shown in a perspective view, with some faces being solid and others dashed to indicate they are behind.

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The diagram shows a 2D foam structure with a red curve and arrows indicating orientation. The foam is divided into two regions by the curve. The top region is shaded with a light blue color, and the bottom region is shaded with a light red color. The red curve is a singular crossing, and the arrows indicate the direction of the differential map.

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with a shift applied to the complex of the negative crossings (but we don't change the sign of the differential).

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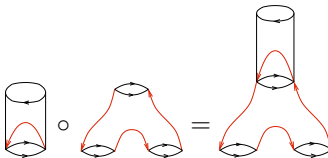
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- As morphisms, we ‘read’ foams from bottom to top and we compose them using vertical concatenation.



Local relations

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$$(S) \quad \begin{array}{|c|} \hline \text{circle} \\ \hline \end{array} = 0, \quad \begin{array}{|c|} \hline \bullet \\ \hline \text{circle} \\ \hline \end{array} = 1$$

$$(UFO) \quad \begin{array}{|c|} \hline \text{foam with red lines} \\ \hline \end{array} = 0 = \begin{array}{|c|} \hline \bullet \\ \hline \text{foam with red lines} \\ \hline \end{array} \quad \begin{array}{|c|} \hline \text{foam with red lines} \\ \hline \end{array} = i = - \begin{array}{|c|} \hline \bullet \\ \hline \text{foam with red lines} \\ \hline \end{array}$$

where $i^2 = -1$.

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where χ is the Euler characteristic, B is the boundary of the web in the source (or target) of the foam F , and d is the number of dots on the foam F .

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- Hence, the differentials in $\llbracket T \rrbracket$ are maps of degree zero.

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- Caprau showed that the construction is an invariant for classical links up to homotopy equivalence, so it is left to check invariance of the Reidemeister type 4 and 5 moves.
- We work with a generating set for the oriented singular Reidemeister moves, but include only two justifications on the next slide.

Invariance under the moves $R4$ and $R5$

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$$\left[\begin{array}{c} \nearrow \quad \nwarrow \\ \bullet \\ \swarrow \quad \searrow \end{array} \right] = \left[\begin{array}{c} \nearrow \quad \nwarrow \\ \times \\ \swarrow \quad \searrow \end{array} \right] \oplus \left(\left[\begin{array}{c} \nearrow \quad \nwarrow \\ \times \\ \swarrow \quad \searrow \end{array} \right] [-1] \right)$$

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$$\begin{aligned}
 \left[\begin{array}{c} \text{Diagram 1} \end{array} \right] &= \left[\begin{array}{c} \text{Diagram 2} \end{array} \right] \oplus \left(\left[\begin{array}{c} \text{Diagram 3} \end{array} \right] [-1] \right) \\
 &\cong \left[\begin{array}{c} \text{Diagram 4} \end{array} \right] \oplus \left(\left[\begin{array}{c} \text{Diagram 5} \end{array} \right] [-1] \right)
 \end{aligned}$$

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 \left[\begin{array}{c} \diagup \quad \diagdown \\ \bullet \\ \diagdown \quad \diagup \end{array} \right] &= \left[\begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array} \right] \oplus \left(\left[\begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array} \right] [-1] \right) \\
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Theorem

The isomorphism class of the homology $\mathcal{H}(L) := H(\mathcal{F}(\llbracket T \rrbracket))$ is a *bigraded invariant* of the singular link L with diagram T , whose graded Euler characteristic equals the polynomial $\overline{P}_L(q)$.

Moreover, if L is a classical link and $a = 0 = h$, $\mathcal{H}(L) \cong Kh(L)$, where $Kh(L)$ is the Khovanov homology of L .

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Thank You!

Questions?

imartinez00@mail.fresnostate.edu